

Abubakr Shafique, Ph.D.

📍 Greater Toronto Area, Ontario, Canada

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Results-driven AI and machine learning researcher with a Ph.D. in Systems Design Engineering from the University of Waterloo, specializing in biomedical image analysis, digital pathology, and representation learning. Experienced in foundation models, self-supervised and multimodal learning, image retrieval, domain adaptation, and deploying clinically relevant AI systems. Combines strong interdisciplinary research, collaboration, mentorship, and academic service to advance computational methods in medicine and biology.

Education

Doctor of Philosophy (PhD), Systems Design Engineering Sep 2020 – Jan 2024

University of Waterloo, Waterloo, ON, Canada

Thesis: Representation Learning for Image Search in Histopathology

Visiting Graduate Scholar Jan 2023 – Jan 2024

Mayo Clinic Graduate School of Biomedical Sciences, Mayo Clinic, Rochester, MN, USA

Department of Artificial Intelligence Informatics

Master of Science (MSc), Biomedical Engineering Feb 2017 – Jan 2019

National Taiwan University of Science and Technology, Taipei, Taiwan

Thesis: A Real-time Non-contact based Adaptive Breathing Monitoring System

Bachelor of Science (BSc), Electrical Computer Engineering Feb 2012 – Jan 2016

COMSATS Institute of Information Technology, Islamabad, Pakistan

Thesis: Parameterized Architecture Design of SENSE MRI Reconstruction Algorithm

Exchange Semester, Electrical Computer Engineering Sep 2014 – Feb 2015

Duzce University, Duzce, Turkey

Research Experience

Post-Doc Research Fellow Apr 2024 – Present

Department of Electrical, Computer and Biomedical Engineering,

Toronto Metropolitan University, Toronto, ON, Canada

- Lead research in medical imaging and digital pathology, developing robust and scalable machine learning solutions for clinical and translational applications.
- Evaluate and benchmark foundation models for representation learning, classification, retrieval, and domain generalization.
- Train, adapt, and fine-tune self-supervised foundation models, including the DINO framework.
- Design efficient model architectures for resource-constrained environments, improving inference speed, memory usage, and computational efficiency.
- Collaborate with clinical pathologists and interdisciplinary teams from UHN, OICR, UoT, and SickKids.
- Supervise graduate and undergraduate students and support grant preparation, lectures, and project coordination.

Digital Pathology Image Analyst (Internship)

Aug 2024 – Feb 2025

Department of Computational Science Informatics

Hoffmann-La Roche Limited, Mississauga, ON, Canada

- Conducted research using multi-modal large language models (MLLMs) on publicly available histopathology datasets.
- Fine-tuned and evaluated vision-language models (VLMs) on HPC GPU clusters using LoRA.
- Developed an automated pipeline for in-house data curation from learning modules related to the PDL-1 Roche assay.
- Performed synthetic data generation for vision-language models (VLMs).

Visiting Graduate Scholar

Jan 2023 – Jan 2024

Department of Artificial Intelligence Informatics

Mayo Clinic, Rochester, MN, USA

- Proposed a novel ranking loss (based on search and retrieval) for learning representations that distinguish fine-grained subtypes in image search.
- Integrated tissue images and immunogenomic transcriptomic data for multi-modal image search.
- Introduced Selection of Distinct Morphologies (SDM) to select representative whole-slide image patches.
- Proposed a compact five-block Vision Transformer for rotation-agnostic image representation learning in digital pathology.
- Benchmarked foundation models and histopathological search engines from the literature.
- Conducted analysis of 35 breast cancer subtypes using WHO data to compare deep and morphological features.

Research Assistant

Sep 2020 – Jan 2024

Kimia Lab, Systems Design Engineering,

University of Waterloo, Waterloo, ON, Canada

- Addressed class-imbalanced unsupervised and semi-supervised domain adaptation for histopathology images.
- Designed Composite Biomarker Image, a visualization method that integrates multiple IHC biomarker images using fuzzy inference systems.
- Proposed an immunohistochemistry biomarkers-guided image search framework for whole-slide histopathology images.
- Developed an automated fine registration method for multiple-stain whole-slide images.

Engineering Computing Consultant

Aug 2021 – Dec 2022

Faculty of Engineering,

University of Waterloo, Waterloo, ON, Canada

- Supported students with remote connectivity, software installation, license access, and device troubleshooting during the pandemic.

Teaching Assistant

Jan 2021 – Dec 2022

Department of Systems Design Engineering,

University of Waterloo, Waterloo, ON, Canada

- Supported laboratory experiments, tutorials, one-on-one sessions, software demonstrations, office

hours, term projects, and grading.

Research Assistant

Feb 2017 – Aug 2020

NTUST Center of Computer Vision and Medical Imaging,

National Taiwan University of Science and Technology, Taipei, Taiwan

- Developed a real-time non-contact adaptive breathing monitoring system using smartphone live camera input.
- Research in deep learning for digital pathology and tools for faster, semi-automatic delineation.

Research Assistant

Feb 2015 – Jan 2016

Medical Image Processing Research Group,

COMSATS Institute of Information Technology, Islamabad, Pakistan

- Developed an ASIC architecture for real-time SENSE reconstruction in pMRI.
- Implemented an FPGA version of the architecture for real-time SENSE MRI reconstruction.

Technical Skills

• Programming Languages

Python, C, C++, C#, CUDA, MATLAB, HTML, CSS, Java, JavaScript.

• Artificial Intelligence / Machine Learning / Deep Learning

PyTorch, TensorFlow, Keras, Scikit-Learn, SciPy, Foundation Models (FM), Self-Supervised Learning (SSL), Multi-Modal AI, Convolutional Neural Networks (CNN), Vision Transformers (ViT), Large Language Models (LLMs), Multi-Modal Large Language Models (MLLMs), Multi-Instance Learning (MIL), Retrieval-Augmented Generation (RAG).

• Computer Vision

OpenCV, ImageJ/Fiji, Scikit-Image, Pillow, Digital Pathology, Biomedical Image Analysis, Image Retrieval, Search Engine, Vector Database, Domain Adaptation.

• Data/Tools

Pandas, Data Analysis, Data Curation, Weights & Biases (W&B), HPC GPU clusters, Jupyter Notebooks.

• Hardware

FPGA, Verilog, VHDL, ModelSim, Xilinx, OrCAD, LTspice, PSpice, Simulink, Proteus.

Awards and Honors

- Awarded Best Poster Presentation for MICCAI 2025 FAIMI Workshop.
- Accepted as a Visiting Graduate Scholar at Mayo Clinic, Rochester, MN, USA.
- International Doctoral Student Award (IDSA), University of Waterloo, Waterloo, ON, Canada.
- Graduate Research Studentship (GRS), University of Waterloo, Waterloo, ON, Canada.
- Awarded a Full Scholarship for MSc in biomedical engineering at the National Taiwan University of Science and Technology, Taiwan.
- Awarded scholarship from COMSATS Institute of Information Technology, Islamabad, Pakistan, to present research paper in European Society for Magnetic Resonance in Medicine and Biology (ESMRMB) 2015 conference in Edinburgh, United Kingdom.
- Awarded a scholarship to study the fall 2014 semester at Duzce University, Turkey, as a visiting scholar.

Invited Talks

- AI to Improve Cancer Care, Canadian Cancer Trials Group Annual Spring Meeting, April 2025.
- How Fair are Foundation Models? Exploring the Role of Covariate Bias in Histopathology, Stanford MedAI Group, October 2025.
- Investigating Scanner-bias in Histopathology Image Retrieval Using Foundation Models, SPIE Medical Imaging, February 2026.
- How Fair are Foundation Models? Exploring the Role of Covariate Bias in Histopathology, Abbvie CVRT Imaging Seminar, February 2026.

Certifications

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| • The University of Edinburgh
<i>Data Science in Stratified Healthcare and Precision Medicine</i> | Oct 2025 |
| • The Johns Hopkins University
<i>Cancer Biology</i> | Oct 2025 |
| • The Johns Hopkins University
<i>Introduction to Neurohacking in R</i> | Oct 2025 |
| • Intel Corporation
<i>An Introduction to Practical Deep Learning</i> | Dec 2020 |
| • deeplearning.ai
<i>Neural Networks and Deep Learning</i> | Aug 2020 |
| • The Johns Hopkins University
<i>COVID-19 Contact Tracing</i> | Jul 2020 |

Academic Service

- Reviewer: Journal of Imaging Informatics in Medicine.
- Reviewer: npj Precision Oncology.
- Reviewer: npj Digital Medicine.
- Reviewer: Surgical and Experimental Pathology.
- Reviewer: Scientific Reports.
- Reviewer: Discover Computing.
- Reviewer: Diagnostic Pathology.
- Reviewer: Communications Medicine.
- Reviewer: Medical Image Analysis.
- Reviewer: Scientific Data.
- Reviewer: BMC Medical Imaging.